



MSc Cyber Security Engineering

Managing Cyber Risk, Audit and Compliance

Risk Management Report

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I hereby declare that this assessment is my own work and that all sources have been acknowledged.

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Introduction

The organisation is a major UK vehicle manufacturer undergoing its first formal cyber security audit, despite long-standing reliance on industrial robots, IoT devices, connected vehicle platforms and engineering systems. Cyber incidents could significantly affect safety, production, intellectual property and customer trust.

This report evaluates how established cyber risk management standards and frameworks apply to this combined IT, OT and IoT environment and selects a suitable approach. Using the selected framework, it analyses the risks associated with the ten technologies identified during the audit and presents findings in a risk register with proposed treatments. The report concludes with prioritised recommendations for reducing cyber risk exposure and strengthening ongoing governance.

1 Critical Analysis of Risk Management Frameworks

1.1 Context of Cyber Risk in Automotive Manufacturing

The manufacturer operates a complex technological environment that includes industrial robotics, IoT gateways, smart cameras, thermal sensors, EV chargers, in-vehicle infotainment systems and engineering software. These technologies are interconnected across both IT and OT networks and, in some cases, extend beyond the factory into vehicles and external infrastructure such as charging points. As a result, cyber threats may propagate across multiple systems, potentially affecting both production facilities and vehicles in the field.

Cyber incidents in this environment may lead not only to data breaches and financial losses but also to safety risks, production disruption, regulatory non-compliance and damage to brand reputation. These concerns are consistent with wider guidance on industrial control systems and automotive cybersecurity (NIST, 2015; ENISA, 2021). Long equipment lifecycles, complex supply chains, and reliance on vendor firmware and network connectivity further increase exposure, particularly where security has not been systematically evaluated since deployment. This mix of safety-critical operations, connected products and legacy assets means the manufacturer needs a structured, repeatable way to manage cyber risk.

1.2 Overview of Selected Risk Management Frameworks

Several established standards and frameworks guide cyber risk management in complex technological environments. This section briefly outlines four widely recognised frameworks relevant to the manufacturer's environment: **ISO 31000** (ISO, 2018a), **ISO/IEC 27005** (ISO, 2018b), **ISO/SAE 21434** (ISO/SAE, 2021) and **IEC 62443** (IEC, 2013-2021). These standards address risk from different perspectives and provide reference points for a mixed IT, OT and connected vehicle context.

The **ISO 31000** standard provides high-level guidance for managing risk across organisations. It establishes a set of principles intended to ensure that risk management is integrated into

governance, decision-making and organisational processes. The framework promotes a structured approach consisting of establishing context, identifying risks, analysing and evaluating them, and implementing treatment and monitoring measures. Because it applies to a wide range of organisational risks, it is commonly used to support enterprise-level risk management strategies.

ISO/IEC 27005 focuses specifically on information security risk management and supports the implementation of the **ISO/IEC 27001** information security management system. The standard provides detailed guidance on identifying information assets, analysing threats and vulnerabilities, and assessing potential impacts on confidentiality, integrity and availability. It is widely used when evaluating cyber risks affecting information systems and digital services, and its methodology can also be extended to operational technology and IoT environments where digital communication and data flows play a critical role.

Domain-specific guidance is provided by standards developed for automotive and industrial systems. **ISO/SAE 21434** focuses on cybersecurity engineering within road vehicles and addresses cyber risk across the entire vehicle lifecycle, including development, production, operation and maintenance. It supports the identification and mitigation of risks affecting in-vehicle systems and connected services such as telematics and infotainment platforms. **IEC 62443** is particularly relevant for factory equipment such as robotic controllers, cameras, sensors and chargers, introducing concepts such as zones, conduits and defence-in-depth.

1.3 Critical Evaluation of Framework Suitability

The four frameworks differ in scope, depth and practicality for the manufacturer's situation. ISO 31000 offers the broadest view, covering strategic, operational, financial, safety and cyber risks in a single enterprise risk process, which fits a vehicle manufacturer where cyber incidents can affect both production and products. However, it is deliberately high-level and does not provide detailed cyber-specific guidance, so it must be tailored and supplemented to address technical risks in IT, OT and vehicle systems.

ISO/IEC 27005 provides a rigorous method for identifying assets, threats, vulnerabilities and impacts, but its focus on information assets means safety and OT consequences can be under-represented if it is not adapted. Its focus is primarily on information assets and confidentiality, integrity and availability, so if used alone it may under-represent safety and operational consequences associated with OT failures and industrial accidents. The manufacturer therefore needs to extend it to factor in safety and production impact when assessing OT and IoT components.

IEC 62443 and ISO/SAE 21434 add important domain-specific depth. IEC 62443 aligns well with the factory's industrial equipment (robot controllers, cameras, sensors, EV chargers) through concepts such as zones, conduits and defence-in-depth, but full implementation can be engineering-heavy, complex and dependent on vendor capabilities and support. ISO/SAE 21434 is tailored to vehicle cybersecurity, telematics and IVI and supports lifecycle-based risk

assessment for connected vehicles, yet it is primarily product-focused and does not by itself provide a factory-wide or enterprise governance perspective. Given this is the manufacturer's first formal cyber security audit, relying solely on IEC 62443 or ISO/SAE 21434 would likely over-emphasise technical controls in specific domains and not give enough attention to governance, prioritisation and coordination across IT, OT and vehicle functions.

1.4 Justification of Chosen Framework Approach

Given the mixed IT, OT and vehicle environment, ISO 31000 is chosen as the overarching framework because it supports enterprise-wide governance and integrates safety, operational and cyber risks in a way that IEC 62443 and ISO/SAE 21434 do not when used alone. ISO/IEC 27005 then provides the main method for identifying assets, threats, vulnerabilities and impacts, enabling a structured risk register and treatment plan. IEC 62443 and ISO/SAE 21434 are applied as domain-specific references for industrial automation and vehicle systems so that technical controls reflect recognised good practice without fragmenting overall governance.

Framework	Strength in this manufacturer's context	Key limitation in this context
ISO 31000	Enterprise-wide governance; integrates safety, operational and cyber risk into one process	High-level; requires tailoring to provide concrete cyber guidance
ISO/IEC 27005	Detailed method for assets, threats, vulnerabilities and a risk register	Focuses on information assets; less explicit on physical safety/OT
IEC 62443	Strong for OT zones, conduits and defence-in-depth in factory environments	Complex, engineering-heavy; limited enterprise governance view
ISO/SAE 21434	Tailored to vehicle lifecycle, telematics and IVI	Primarily product/vehicle focused; does not cover factory-wide risk

Table 1.4.1 summarises the specific strengths and limitations of each framework in the context of this manufacturer.

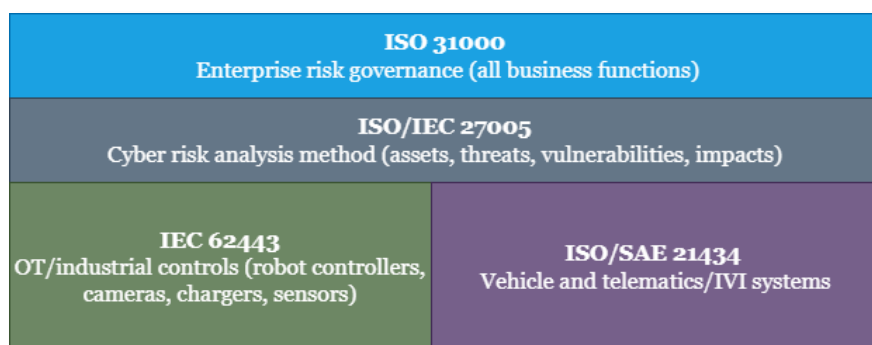


Figure 1.4.1: Layered use of ISO 31000, ISO/IEC 27005, IEC 62443 and ISO/SAE 21434 across the manufacturer's environment.

2 Risk Analysis of Key Technologies

2.1 Risk Assessment Methodology

This section applies a combined approach based on ISO 31000 and ISO/IEC 27005 to assess the cyber risks associated with the ten technologies identified during the audit. In line with the principles of ISO 31000, the assessment follows a structured process that includes establishing context, identifying risks, analysing and evaluating them, and proposing appropriate treatment measures (ISO, 2018a). The methodology described in ISO/IEC 27005 is used to guide the identification of assets, threats, vulnerabilities and potential impacts on confidentiality, integrity and availability within the manufacturer's interconnected IT, OT and IoT environment (ISO, 2018b).

A qualitative risk scoring approach is adopted using five-point scales for likelihood (1 = rare, 5 = almost certain) and impact (1 = insignificant, 5 = catastrophic). Likelihood considers exposure to external networks, known vulnerabilities and ease of exploitation; impact considers consequences for safety, production continuity, confidentiality, regulatory compliance and reputation. Where available, publicly disclosed CVEs and vendor security advisories (for example for ABB RobotWare, Volkswagen MIB3, Siemens VersiCharge, Axis and FLIR devices) are used as evidence when judging exploitability and potential impact. Overall risk levels combine likelihood and impact to categorise risks as low, medium or high, providing a structured basis for prioritising treatment.

The assessment is based on several assumptions derived from the audit scenario. It is assumed that the identified technologies are deployed in configurations typical of a large automotive manufacturer and that known vulnerabilities have not yet been fully remediated. These assumptions support consistent, comparable risk ratings across technologies and reflect the organisation's current cyber security maturity

2.2 Risk Register for Identified Technologies

Asset / Technology (reference)	Risk description	Threat and vulnerability	Likelihood (1-5)	Impact (1-5)	Risk level	Existing controls (assumed)	Proposed treatment (summary)
ABB robot controllers (ABB, 2024; ABB, n.d.)	Unauthorised changes to robot programs causing unsafe movements, defective output and large-scale re-work	Attacker or malicious insider exploits weak access control, flat OT network or unpatched RobotWare firmware (CVE-2024-1913) to change controller configuration	4	5	High	Basic OT network, vendor defaults largely in place, limited role-based access	Segment OT network (zones/conduits), harden access (MFA, strong auth), restrict remote access via jump hosts, enforce change control and logging, apply security updates promptly
KUKA robot controllers (KUKA, n.d.)	Unauthorised changes to robot programs causing unsafe movements, defective output	Attacker or malicious insider exploits weak access control, flat OT network or	4	5	High	Similar OT network design, vendor defaults,	As for ABB: OT segmentation, hardened access, managed remote access, strict change control

	and large-scale re-work	unpatched controller firmware to change configuration				limited role separation	and logging, formal firmware/patch process
Cognex 3D smart camera (Acrovision, 2022)	Manipulated or disrupted inspection data leading to undetected defects	Attacker accesses camera interface over insecure network and alters configuration or feed; weak authentication, no integrity check	3	4	High	Standard IP security, some password changes, basic physical security	Isolate cameras on OT VLANs, enforce strong authentication, encrypt management traffic, enable logging, perform periodic configuration reviews and validation checks in QC
AutoCAD / engineering software (Autodesk, 2023)	Theft or tampering of design files causing IP loss or compromised designs	Phishing, malware or unauthorised access to engineering workstations/file shares; insufficient access control and endpoint protection.	3	5	High	Corporate AV, AD-based access control, regular backups	Enforce least-privilege on design repositories, strengthen endpoints (EDR), strict patching, version control and integrity checks, DLP for sensitive design data
FANUC virtual robot controller (FANUC, n.d.)	Compromised simulation environment propagating malicious logic into production robots	Attacker modifies simulation programs/configurations later deployed to live robots; weak separation between simulation and production networks	3	4	High	Some IT/OT segregation, informal program validation	Separate simulation from production networks, require code review and approval, digitally sign and verify programs before deployment, restrict access to simulation servers
Syrus4 IoT gateway (telematics) (DCT, 2023)	Unauthorised access to telematics data and remote manipulation of telemetry exposing fleet-wide location/driver data	Remote attacker exploits exposed interfaces or weak API security; inadequate encryption or authentication on data channels.	4	4	High	Basic APN/VPN, SIM-based access control, perimeter firewalls	Enforce strong mutual authentication and encryption, harden device configuration, restrict management interfaces, add anomaly detection on telematics traffic, run regular vendor-supported tests
Volkswagen MIB3 IVI unit (PCA Cybersecurity, 2025; Volkswagen AG, n.d.)	Compromised IVI enabling access to personal data or lateral movement to safety-critical vehicle networks and regulatory non-compliance	Exploitation of IVI vulnerabilities (Bluetooth/media issues such as CVE-2023-28910, CVE-2023-28902) and insecure interfaces; weak isolation between IVI and other	3	5	High	OEM security features, some in-vehicle segregation, standard update process	Ensure strict separation between IVI and safety-critical networks, implement secure update mechanisms, minimise data collection/retention, perform regular penetration

		in-vehicle networks					testing aligned with ISO/SAE 21434
Siemens VersiCharge AC chargers (Siemens, 2023)	Remote disruption of charging causing operational downtime or safety incidents	Exploitation of networked management or Modbus services (including vulnerabilities disclosed in 2025 such as <i>CVE-2025-31930</i>) due to weak authentication and insecure defaults, enabling remote charger control.	3	4	High	Basic network security, vendor cloud management, physical protection	Segment chargers on dedicated networks, enforce strong authentication on management interfaces, disable unnecessary services, define firmware update process, monitor charger logs for anomalies
Axis PTZ cameras (Axis Communications, n.d.)	Loss of visibility or privacy breaches due to camera hijacking	Attacker gains access to streams/configuration via exposed interfaces, default/weak credentials or camera firmware flaws (Axis Q6128-E issue <i>CVE-2024-47257</i>)	3	3	Medium	Some password controls, NVR recording, physical security	Harden configs (unique strong passwords, disable unused services), segment on OT VLANs, encrypt management and streaming where possible, restrict remote access, regularly review access logs
FLIR AX8 thermal cameras (FLIR, n.d.)	Disabled or manipulated thermal monitoring causing undetected overheating and equipment damage	Compromise of camera configuration or connectivity through firmware vulnerabilities or weak authentication (<i>CVE-2022-37061</i>) with little monitoring	2	4	Medium	Vendor defaults, integrated with monitoring systems	Secure deployment (segmentation, strong auth), restrict management access, integrate with SIEM, periodically test alerting, keep firmware up to date
All listed technologies (cross-cutting)	Inconsistent patching and configuration management increasing exposure to known vulnerabilities	Complex asset base, fragmented responsibility, no central OT/IoT patch management; reliance on vendor notifications	4	4	High	Ad hoc patching, some IT patch processes, limited OT coverage	Establish centralised asset inventory, define OT/IoT patch and configuration process, agree vendor SLAs, use risk-based patching schedule, apply formal change control and testing

Table 2.2.1 summarises key risks for the ten technologies using the Section 2.1 methodology.

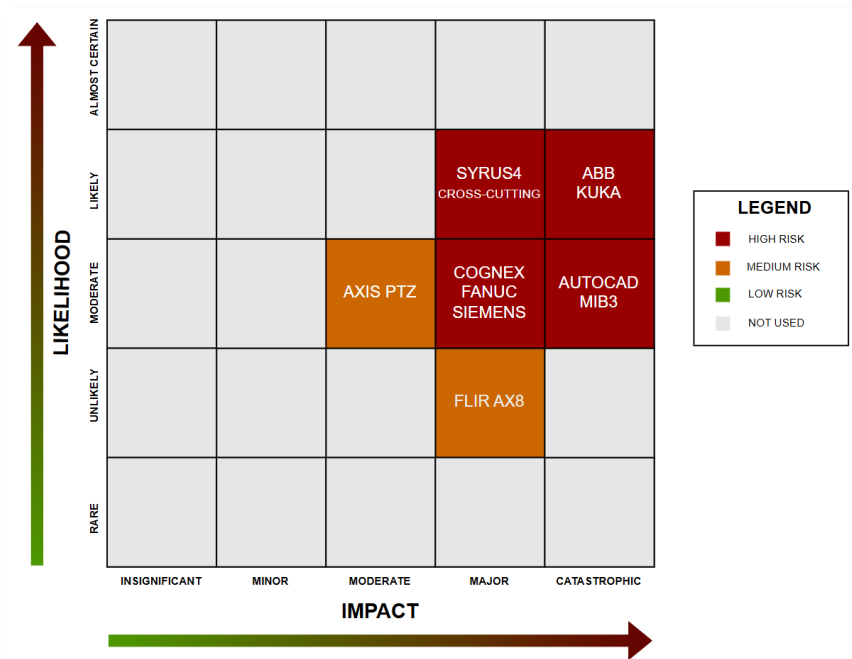


Figure 2.2.1: presents the corresponding likelihood-impact risk matrix, plotting each technology according to the scores in Table 2.2.1.

2.3 Risk Treatment and Prioritisation

The risk assessment identified several technologies presenting the highest cyber risk. Industrial robot controllers (ABB, KUKA, FANUC virtual controller) are high-impact assets due to their direct role in manufacturing; compromise could lead to safety incidents, production disruption and quality issues. The Syrus4 IoT gateway and MIB3 IVI unit present elevated risks because they handle telematics data and support vehicle connectivity; security weaknesses could expose sensitive data, cause regulatory breaches, or influence vehicle behaviour. Siemens VersiCharge chargers carry significant risk because remote compromise could disrupt availability or create safety concerns. These technologies received the highest ratings (impact 4-5, likelihood 3-4) and require immediate attention. Several of these high-impact risks also intersect with safety and data protection obligations, increasing potential regulatory and recall impact.

Impact scores of 5 are used where compromise could directly affect safety and large-scale production or critical intellectual property, such as robot controllers, telematics/IVI platforms and engineering design data. Scores of 4 reflect serious but more contained operational impact, for example disruption of EV charging or quality-control cameras. Likelihood ratings of 3-4 are driven by factors such as internet or cloud connectivity, widespread deployment and known vulnerabilities, while devices with more limited exposure, such as FLIR and some Axis cameras, remain at medium likelihood.

Analysis of the risk register also reveals several recurring control weaknesses across multiple technologies. These themes are addressed in Section 3 recommendations.

In accordance with the principles of ISO 31000, the highest risks should primarily be addressed through risk reduction measures. This includes implementing stronger technical controls such as improved network segmentation, device hardening and secure remote access mechanisms, alongside governance improvements such as clearer change management processes and enhanced monitoring of critical assets. The prioritised treatment actions identified here form the basis for the strategic and operational recommendations outlined in Section 3.

3 Recommendations for Risk Reduction

3.1 Overall Cyber Risk Posture

The assessment shows high overall cyber risk exposure driven by safety-critical OT systems (robot controllers, chargers, sensors), connected vehicle technologies (telematics gateway, MIB3 IVI) and critical information assets (engineering designs). The highest-priority risks combine high impact with medium-to-high likelihood, creating significant potential consequences for safety, production continuity, confidentiality and regulatory compliance.

Risk patterns across the ten technologies highlight structural weaknesses, particularly limited segmentation between IT, OT and vehicle networks, weakly governed remote access to OT/IoT assets, fragmented patch and configuration management, and minimal centralised monitoring. These are not isolated technical gaps but systemic issues that cut across sites, technologies and teams. Addressing them requires coordinated governance, architectural change and improved operational discipline rather than device-by-device fixes.

Recommendations in Section 3.2 focus on establishing an enterprise cyber risk process, strengthening governance of OT/IoT assets and embedding continuous improvement. Section 3.3 then sets out how reporting and feedback loops can sustain this posture over time.

3.2 Strategic and Governance Recommendations

The following recommendations are grouped into governance, technical and architectural controls and operational measures. Each item identifies the primary risks addressed, a responsible stakeholder, an indicative timeframe and the expected benefit, to support clear ownership and prioritisation.

Recommendation	Description	Related risks	Responsible stakeholder	Priority / timeframe	Expected benefit
Establish ISO 31000-aligned cyber risk process	Define and document a cyber risk management policy and process covering IT, OT and vehicle systems.	All High risks; cross-cutting gaps	CISO, Risk Manager	High, 0-6 months	Consistent risk identification, prioritisation and reporting
Maintain integrated cyber risk register	Use an ISO/IEC 27005-based register with named owners	All risks in Section 2	CISO, Risk Committee	High, 0-6 months	Improved visibility and

	and regular review cycles.				accountability for risks
Implement OT network segmentation (zones/conduits)	Separate IT, OT and vehicle-related networks using IEC 62443 zones and conduits.	Robots, cameras, chargers, telematics, IVI	OT Network Architect, Plant IT	High, 0-12 months	Reduced lateral movement and impact of compromises
Harden remote access to OT/IoT	Standardise secure remote access (VPN, MFA, jump hosts) and remove unmanaged paths.	Controllers, chargers, cameras, telematics	CISO, OT Ops	High, 0-9 months	Lower likelihood of unauthorised access and changes
Formal OT/IoT patch and config management	Maintain asset inventory, define patch/change process and vendor SLAs for OT/IoT.	Cross-cutting vulnerability risk	OT Engineering, Vendors, IT	High, 0-12 months	Reduced exposure to known vulnerabilities
Strengthen OT/IoT logging and monitoring	Integrate key OT/IoT assets with central logging/SIEM and alert rules.	All high-impact OT/IoT risks	SOC, OT Ops	Medium/High, 0-12 months	Faster detection and response to anomalies
Apply automotive cybersecurity to telematics and IVI	Align Syrus4 and MIB3 security with ISO/SAE 21434 (segregation, secure updates, data minimisation, testing).	Telematics, IVI risks	Vehicle Cybersecurity Lead, OEMs	High, 0-18 months	Reduced privacy and safety risk in connected vehicles
Protect engineering design assets	Enforce least-privilege, version control and integrity checks; deploy EDR/DLP on engineering endpoints.	AutoCAD / design risks	Engineering Manager, IT Security	Medium/High, 0-12 months	Lower risk of IP theft or design tampering
OT/IoT security awareness and training	Train OT engineers and operators on secure operation, change control and incident reporting.	Most OT/IoT risks	HR, CISO, Plant Manager	Medium, 0-12 months	Fewer configuration errors, better incident handling

Table 3.2.1 summarises the key recommendations from the risk analysis, grouped into governance, technical and operational measures, with each linked to specific risks and assigned owners, priorities and timeframes.

These actions target the highest-priority risks identified in Section 2 and the cross-cutting weaknesses around segmentation, remote access, patching and monitoring. Governance measures provide a consistent framework for decision-making, while technical and

operational controls improve the security and resilience of the most critical technologies in both factory and vehicle environments.

3.3 Closing the Loop: Reporting and Continuous Improvement

To sustain risk reduction, the manufacturer should embed cyber risk reporting and continuous improvement into its governance arrangements. An integrated risk register based on ISO 31000 and ISO/IEC 27005 should be reviewed regularly (for example, quarterly), with updates reflecting changes in the threat landscape, incident trends, audit findings and technology deployments.

Reporting should be tailored to key stakeholder groups. Senior management should receive concise summaries of top risks, trends and required investment decisions. Plant operations should see asset-level status on segmentation, patching and monitoring, along with clear actions and playbooks. Vehicle engineering teams should receive focused reports on telematics and IVI risks, testing results and change backlogs

Metrics such as the proportion of critical OT/IoT assets with network segmentation, time to remediate high-risk vulnerabilities and coverage of OT/IoT logging can be used to track progress. As new OT, IoT and vehicle systems are introduced, they should be added to the asset inventory and assessed using the same process, so that cyber risk management stays aligned with operational objectives and evolving regulatory expectations.

4 Conclusion

This report applied a structured risk management approach based on ISO 31000 and ISO/IEC 27005 to assess cyber risks across the manufacturer's mixed IT, OT and connected vehicle environment. Analysis of ten key technologies highlighted significant exposures affecting safety-critical systems, connected vehicle platforms and engineering assets, with high risks concentrated around robot controllers, telematics, IVI and EV chargers.

The resulting risk register and treatment priorities informed governance, technical and operational recommendations addressing both asset-specific threats and cross-cutting weaknesses in segmentation, remote access, patch management and monitoring. Implementing these measures and embedding regular risk reporting and continuous improvement will progressively reduce cyber risk exposure and support safe, reliable and compliant operations.

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